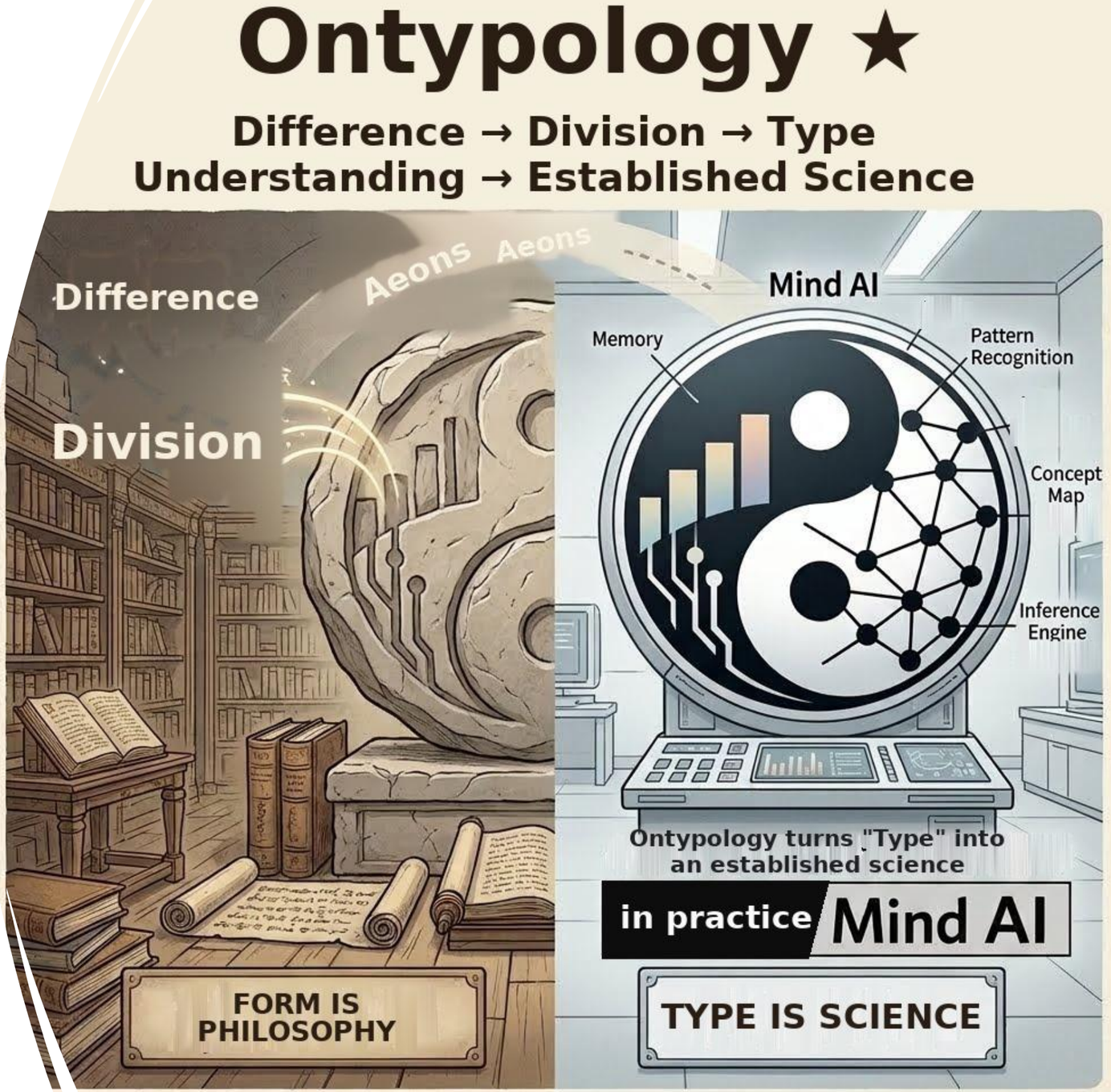
Ontypology

- From difference to division, from division to type, from type to understanding – the brain has walked this path for aeons.
- A century ago, logic was philosophy; once seen clearly, it became an established science.
- Today we explore "Type" (a brain concept) as philosophy; Ontypology makes "Type" clear, turns it into an established science, and puts it into practice as Mind AI.



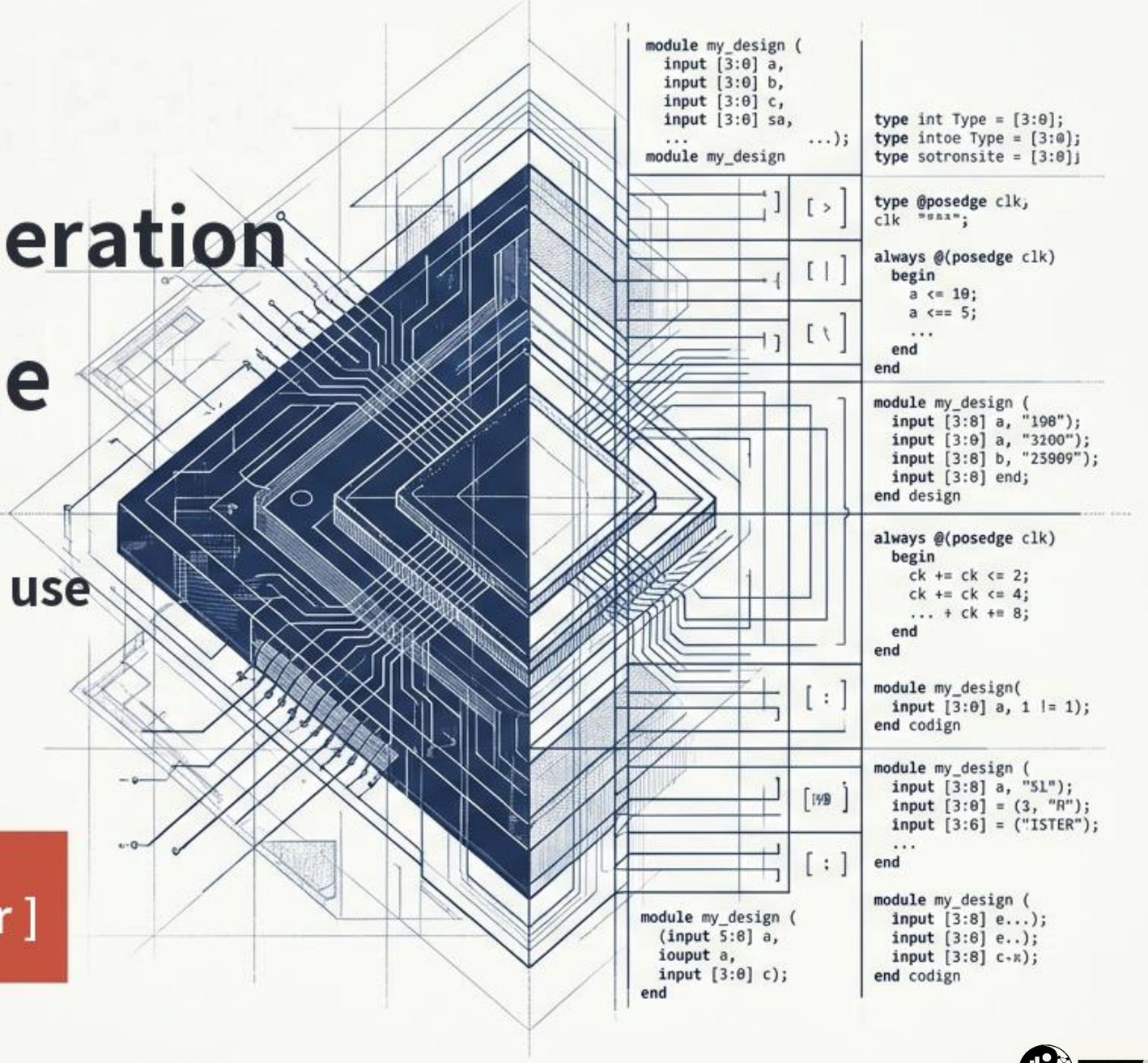
IC Design Brain:

Accurate Verilog Generation

A Type-Driven Guide

From chaos to verifiable truth —
an engineer's guide to accurate AI use

[Core rule: design = Type + Number]



Before we start: three non-negotiable conventions



Convention 1 · Type (not Form)

The English for Type is always Type, never Form. A type is not a catalog or a label; it is a rigorous structure that states its criteria and supports reasoning.



Convention 2 · Theory used, not taught

The Spatial Type's three brackets, the ∩(T) criterion and the three remove levels are applied directly as practical tools; the underlying ontology is not restated.



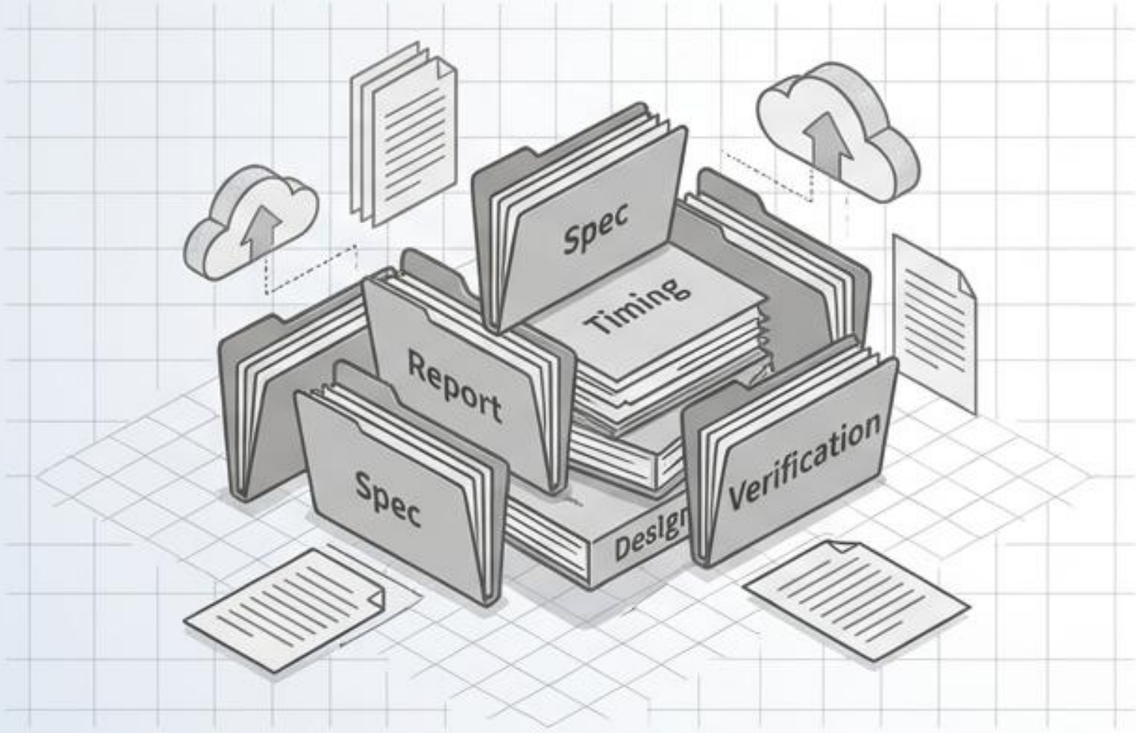
Convention 3 · Platform cited, not described

Mind AI is only the underlying platform that carries the brain. This book focuses on how engineers get IC design right on that platform, not on developing the platform itself.

MIND AI

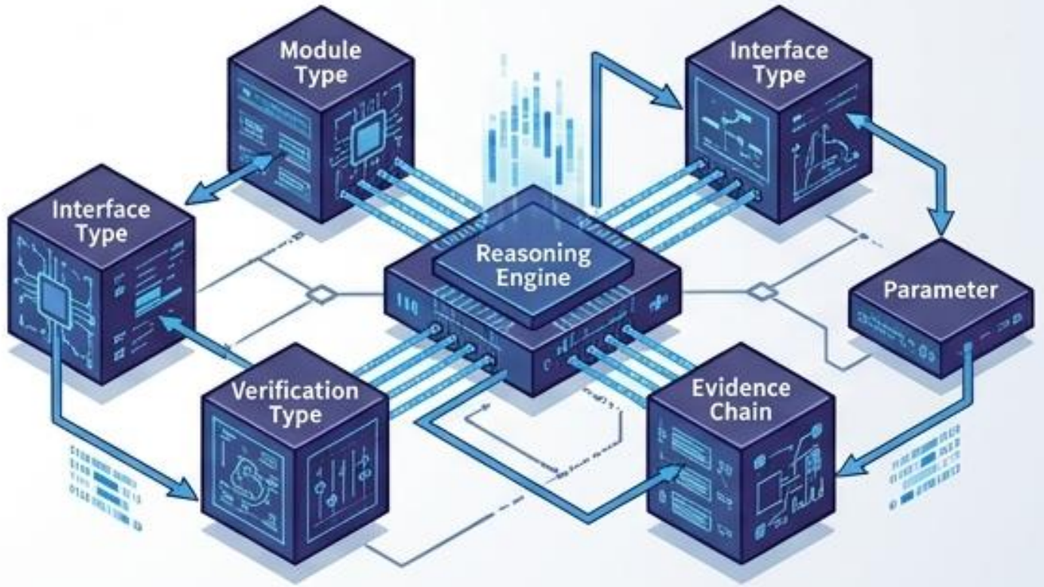
The status quo: a pile of dead files

- Specs live in the cloud; timing diagrams have no version control.
- Verification reports are cut off from the design.
- Files are findable; answers are not.



The brain = a design replica that can reason

- Knowledge: the type warehouse can reason.
- Thinking: the move set can be invoked.
- Evolution: the evidence chain accumulates.
- Context: change 8N1 and the brain lists the affected interface, module and verify types instead of waiting for verification to fail.

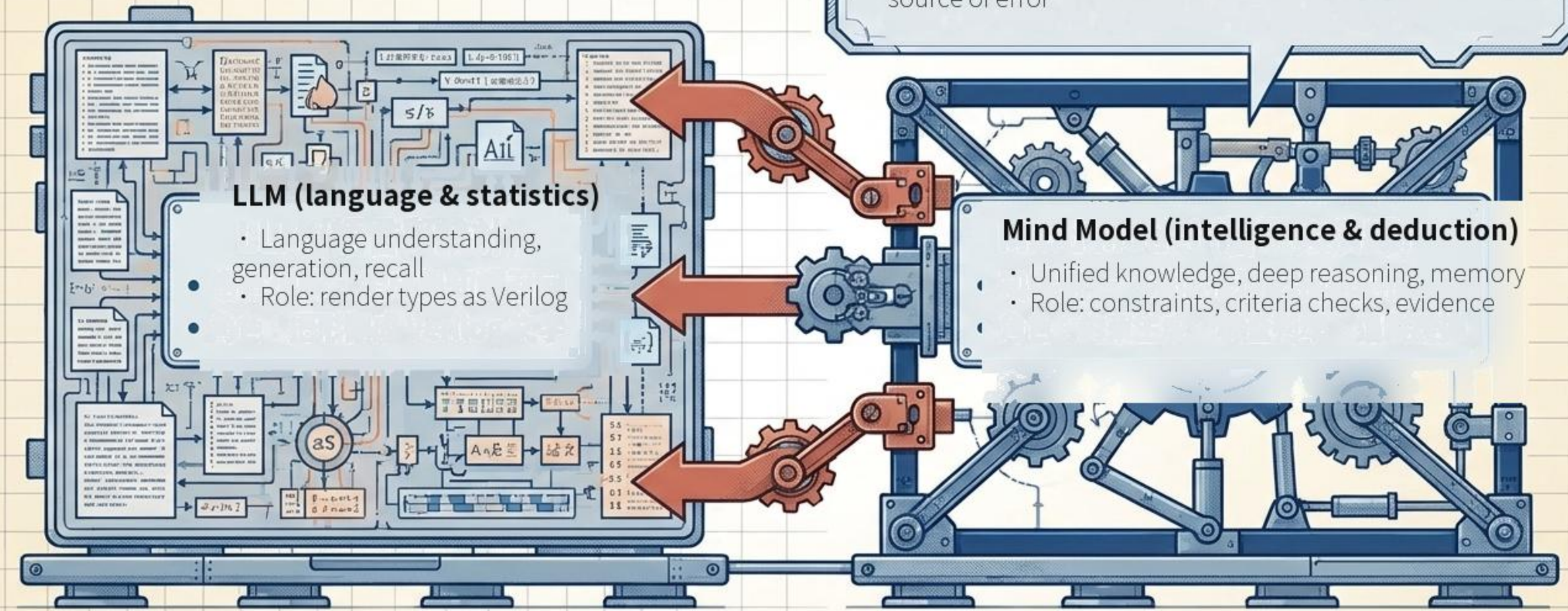


[Axiom] Design = Type + Number. Types first, numbers after. If the type is wrong, even the most precise bit width is meaningless.

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Mind AI architecture: accuracy comes from division of labour

Common myth: longer prompt, higher accuracy
Fact: asking AI to write a whole chip at once is the biggest source of error



LLM (language & statistics)

- Language understanding, generation, recall
- Role: render types as Verilog

Mind Model (intelligence & deduction)

- Unified knowledge, deep reasoning, memory
- Role: constraints, criteria checks, evidence

Models change, but the type and the criteria stay in the brain. Accuracy is not luck; it is the strict constraint of the deductive skeleton.

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The Six Output Types at a Glance

Without types, outputs are loose parts. Every RTL output must hang on one bracket type, and every bracket type carries a criterion ∩(T).

SpecType [ ]

What does it do? (property-led)
Acceptance: every sentence measurable and decidable.

InterfaceType [ ] + ( )

How does it face outward?
Acceptance: it lines up, and it speaks clearly.

ModuleType { }

What is it composed of? (BOM-led)
Acceptance: BOM criteria + remove grading.

TimingType ( )

When does it move? (container-led)
Acceptance: scope-visible, assertion-ready.

VerifyType [ ] { } ( )

How is rightness computed?
Acceptance: coverage met + item-by-item sign-off.

PatternType { } + [ ]

What is reused?
Acceptance: criteria attached + pit records.

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Deep dive: SpecType and InterfaceType

SpecType: health check for specs

Iron rule: unmeasurable adjectives are sent back.

∩(SpecType) checks:

- **Measurable: what number measures it?** ✓ [pass] ✗ [reject]
  (e.g. 115200 bps, 8N1 [✓]
  vs fast transmission [☒])
- **Attributable: which bracket?** ✓ [pass]
  (e.g. 50 MHz fits the ( ) container [✓])
- **Decidable: do two engineers agree?** ☐

InterfaceType: the outward contract

Iron rule: fix direction, bit width and clock domain at once.
Change the interface, change the contract.

UART Port Contract example:

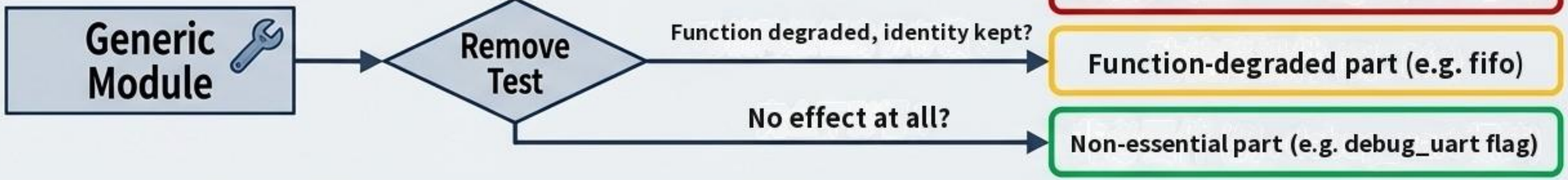
Signal	Direction/Width	Clock domain / attribution
`clk, rst_n`		50 MHz clock domain / Reset domain
`tx_data[7:0], tx_start`		transmit side (50 MHz)
`rx, tx`		Serial port

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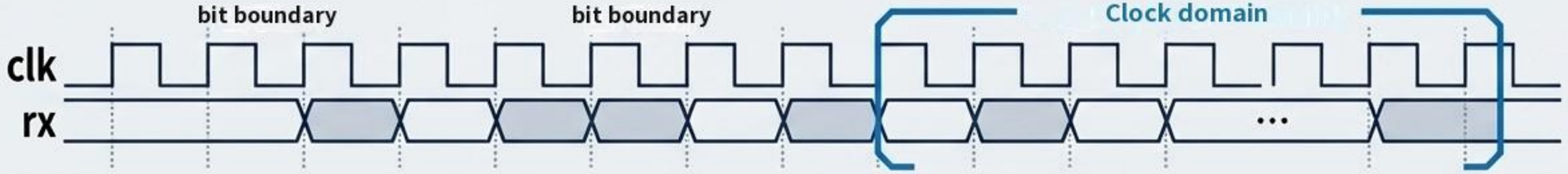
Deep dive: ModuleType and TimingType

ModuleType & the three remove levels

{} BOM records composition. Does removal destroy identity?



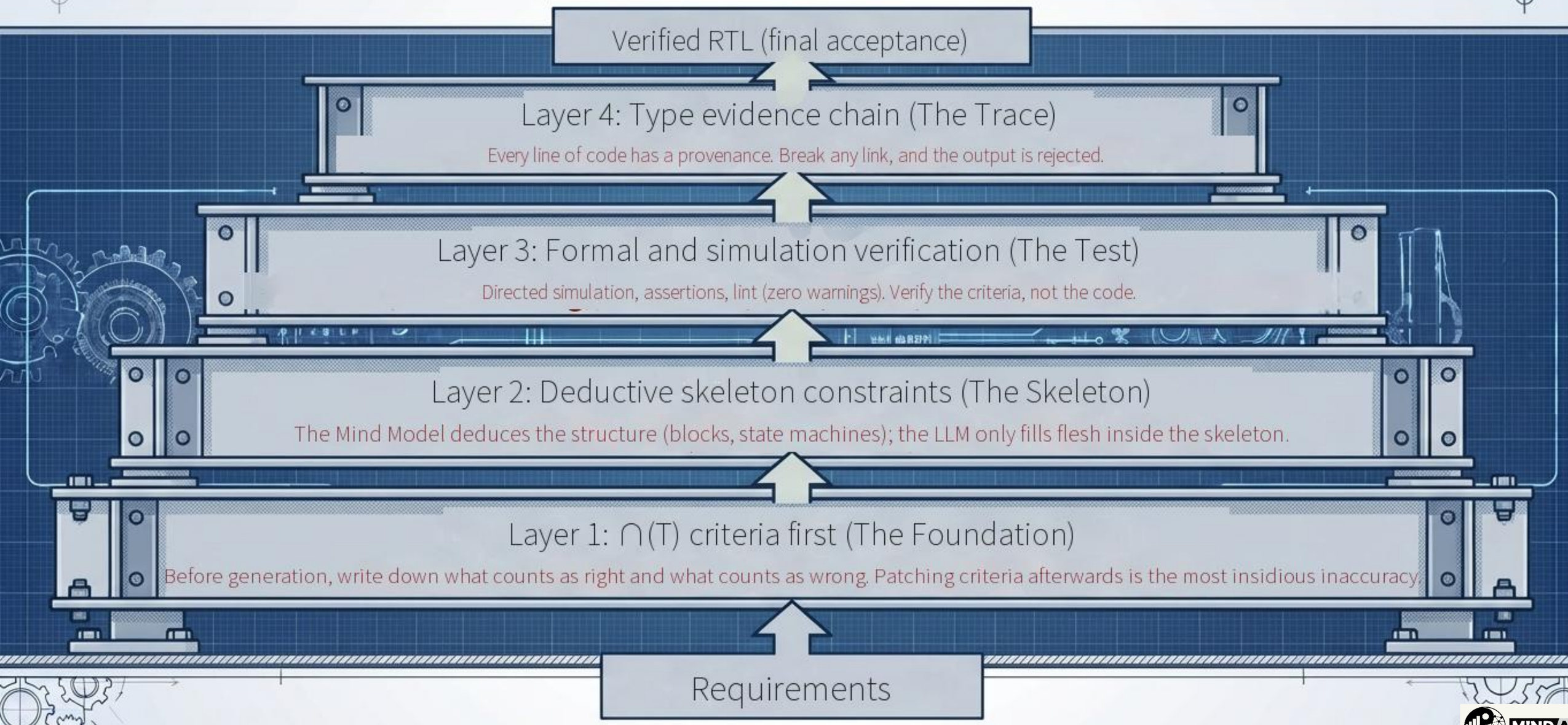
TimingType essentials: the dynamic projection of a ( ) container; criteria always scope-visible.



- Clock spec (SDC)
- Frame timing (baud_tick alignment)
- Reset sequence (polarity/release)
- Cross-domain (CDC synchroniser)

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Four-layer accuracy guarantee: the structured source of trust



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Formal verification and the formal evidence chain: how outputs are accepted

The type evidence chain: the art of tracing

1. Requirements: stop-bit check of the rx module
2. Inference: causal reasoning (temporal native form)
3. Criteria: V-3 frame error rate < 10^{-6} (VerifyType)
4. Evidence: evolution account v1.2, fix record

Right but unable to say why it is right is as good as never verified.

Space × time cross-acceptance matrix

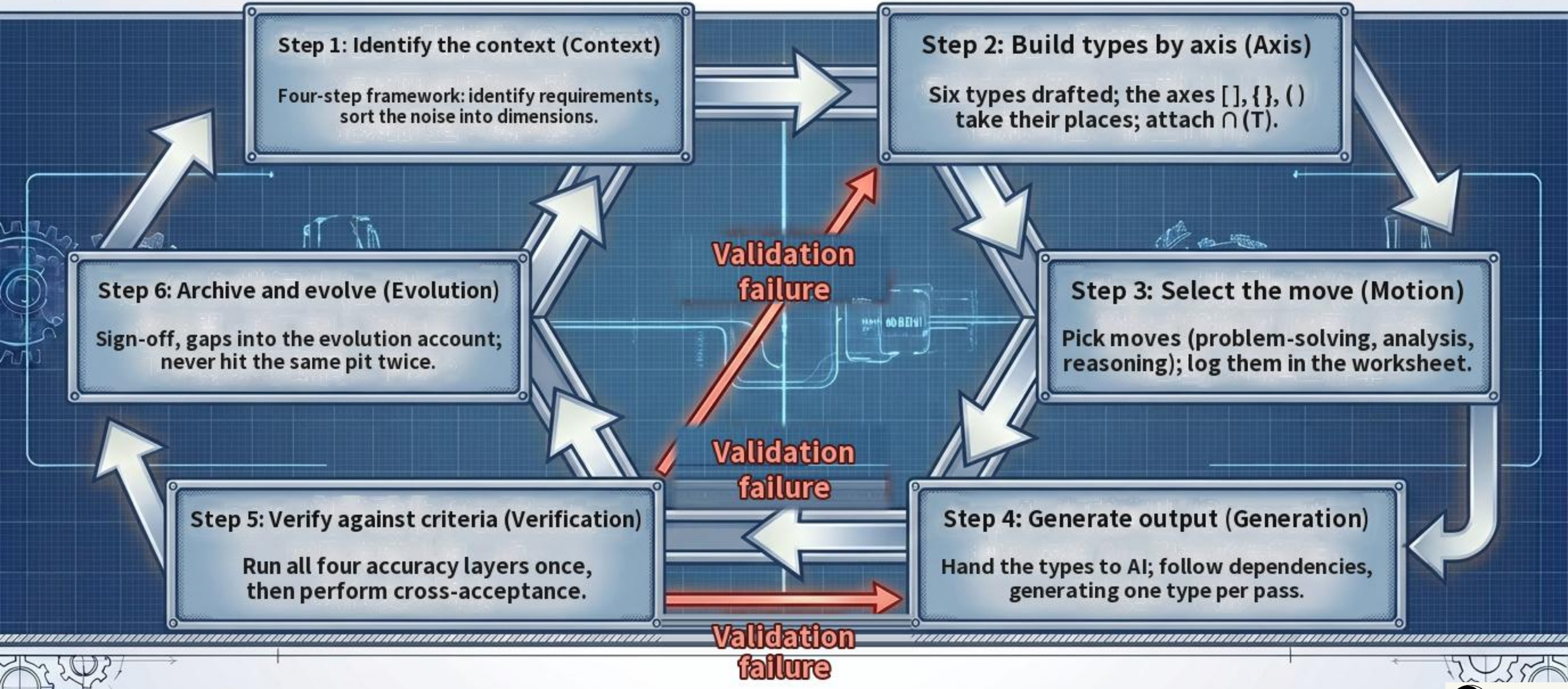
Space (static type): is the form complete?
(all six types present, criteria set, notation complete)



Time (dynamic moves): are the moves in place?
(reasoning complete, records archived)

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Using It Accurately: The Six Steps (The 6-Step SOP)



The Prompt Pipeline S1–S4: From Forms to Skeleton

[AI iron rule: one type per pass]

[S1] Forms

Prompt: List this case's six types and each type's ∩(T) criteria; do not write code.

Output:
Six-form worksheet.

[S2] Spec

Prompt: Write the SpecType from the requirement sentences; each sentence carries an acceptance metric.

Output:
SpecType.

[S3] Interface

Prompt: Define the port list from the SpecType (name, bit width, direction, clock domain).

Output:
InterfaceType.
AI may not add or remove signals on its own.

[S4] Skeleton

Prompt: Give the module skeleton from the InterfaceType (blocks, state machine, drive relations); leave the function empty for now.

Output:
ModuleType.

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The Prompt Pipeline S5–S7: From Fill to Sign-off

[AI iron rule: one type per pass]

[S4] Skeleton

Prompt: Implement the function block by block, one always block at a time, keeping the skeleton structure.

Prompt: Write a self-checking testbench and assertions per ∩(T).

Prompt: Run coverage, list the gaps, summarise the evidence chain.

[S4] Skeleton

[S5] Fill

[S6] Verify

[S7] Sign-off

Output:

Six-type worksheet.

uart_tx Snippet:

```
localparam integer DIV = CLK_FREQ
 / BAUD; // 50e6/115200 = 434
```

SystemVerilog Snippet:

```
!tx_busy |-> tx == 1'b1;
// idle: tx is high
```

Output:

Final sign-off package

(Verified RTL).

Skeleton-layer locked discipline: each register has exactly one driver.

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Five AI Misuses Compared: Escaping the 'Type Ritual' Trap

Most inaccuracy is a type error, not a syntax error: the compiler catches syntax; only accuracy guarantees catch types.

Symptom	Correction
Type as folder vs. mandatory criteria Symptom: six types listed, the work is called done, criteria column blank.	Fix: every type must fill ∩(T); blanks may not enter the flow.
Criteria patched later vs. order locked Symptom: RTL written first, then AI is asked to “think of tests”.	Fix: S1 order locked — criteria are produced before code generation.
Broken evidence chain vs. on-chain spot-check Symptom: verification done, handed over, with no trace back to requirements.	Fix: spot-check the evidence chain; broken ones are rejected.
Numbers before types vs. property arbitration Symptom: the team argues 8 vs. 16 bits first, then thinks about architecture.	Fix: types first, numbers after; numerical disputes go back to [ ] property arbitration.
Interface unlocked vs. contract first Symptom: AI adds signals that nobody asked for.	Fix: S3 contract first — the contract table is locked; no additions or removals.

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Ten Questions Before Acceptance:

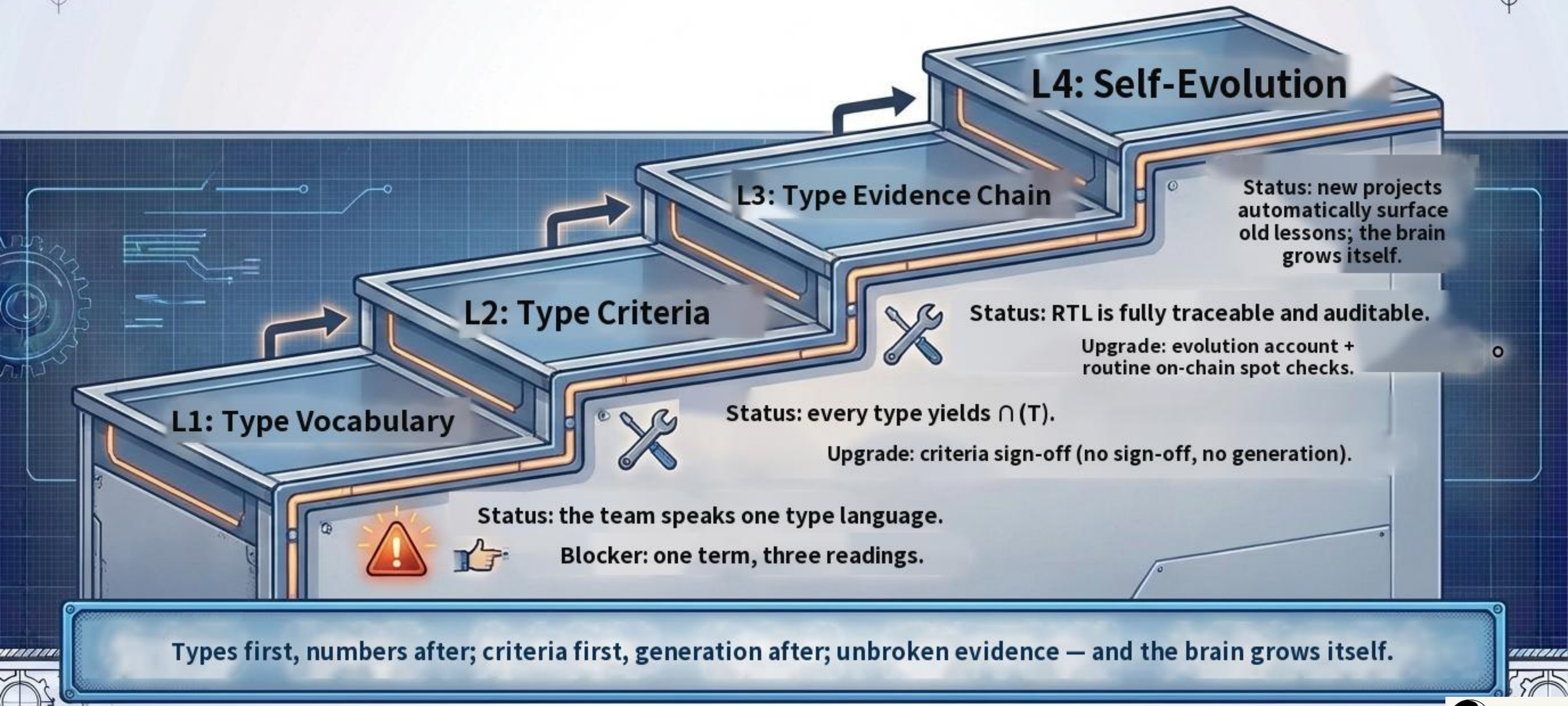
The Last Line of Defence
(The Diagnostic Checklist)

Warning: any "no" = return for rework

- ☐ Does every output type have ∩(T)?
  Evidence: six-type worksheet
  Evidence: sign-off sheet date
- ☐ Were the criteria set before generation?
  Does each always block do one thing?
  Evidence: skeleton document
- ☐ Is the interface contract table locked?
  Evidence: port contract table
  Is the testbench self-checking?
  Evidence: tb source
- ☐ Are the BOM criteria and the remove grading written?
  Evidence: BOM list
  Coverage met, gaps handled?
  Evidence: coverage report
- ☐ Does every cross-domain signal have a synchroniser bridge?
  Evidence: cross-domain list ( )
  Are move invocations and reasoning steps recorded?
  Evidence: worksheet
- ☐ Does each always block do one thing?
  Evidence: skeleton document
  Does the evidence chain trace back to the requirement sentence?
  Evidence: on-chain spot check

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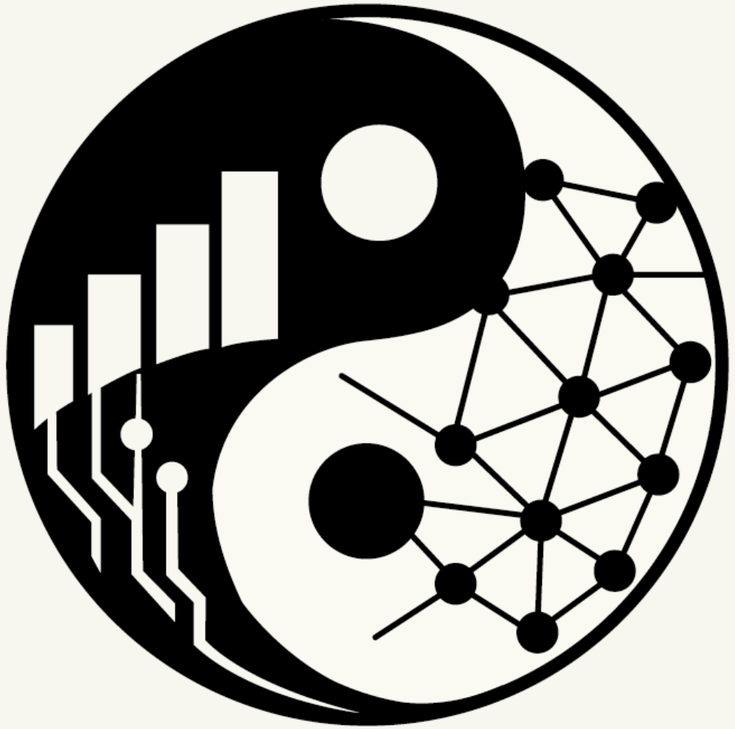
Maturity Ladder (L1–L4): Grow Your Own IC Design Brain



Types first, numbers after; criteria first, generation after; unbroken evidence — and the brain grows itself.

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LonGreat Vision: to create intelligent life that understands the essence of the world



MIND AI

"We aim to make artificial intelligence not merely a tool, but a system of intelligent life that truly possesses the abilities to understand, reason, learn, and create."

From the unification of knowledge, to the unification of thought, and finally toward the unification of intelligence – this is the eternal path to AGI that LonGreat pursues.

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